

Report: Basic Auditing & Logging

1. Introduction

Basic auditing and logging settings were configured to enable visibility into key security-related activities within the Active Directory environment. These tasks focused on capturing user logon events, monitoring access to specific files and folders, and enforcing account lockout policies after repeated failed login attempts. Group policies were created, linked to the appropriate Organizational Units, and configured with specific auditing rules. The effectiveness of these policies was validated on a domain-joined client machine using Event Viewer and administrative tools. The logon, object access, and account lockout events were successfully captured and verified, ensuring that foundational auditing controls are in place for security monitoring and future incident analysis.

2. Objective

The objective of these tasks was to establish basic auditing and logging capabilities in a domain environment by configuring group policies for logon event auditing, object access tracking, and account lockout control. This included creating and linking dedicated GPOs, enabling relevant audit settings under Audit Policy, and applying folder-level auditing on a test directory. The logon event auditing was tested by signing in with a domain user account and confirming Event ID 4624. Object access auditing was validated by interacting with an audited folder and verifying Event ID 4663. The account lockout policy was enforced by configuring threshold and duration values in the Default Domain Policy, then triggering a lockout through multiple failed login attempts. All activities were monitored and confirmed using ADUC and Event Viewer.

3. Tools used

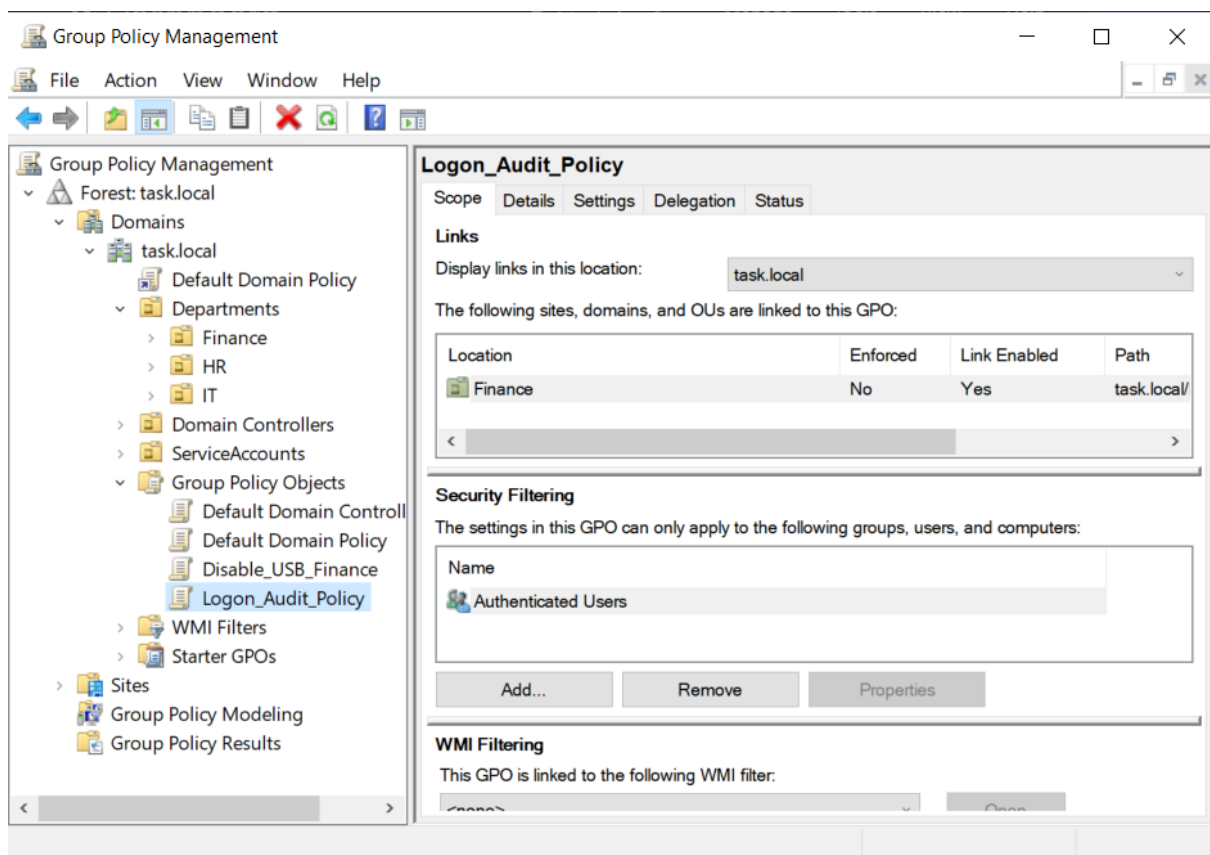
- Windows Server (AD DS)
- Windows PowerShell

- Group Policy Management Console (GPMC)
- Event Viewer
- Active Directory Users and Computers (ADUC)
- Windows 10 Enterprise (PC1)

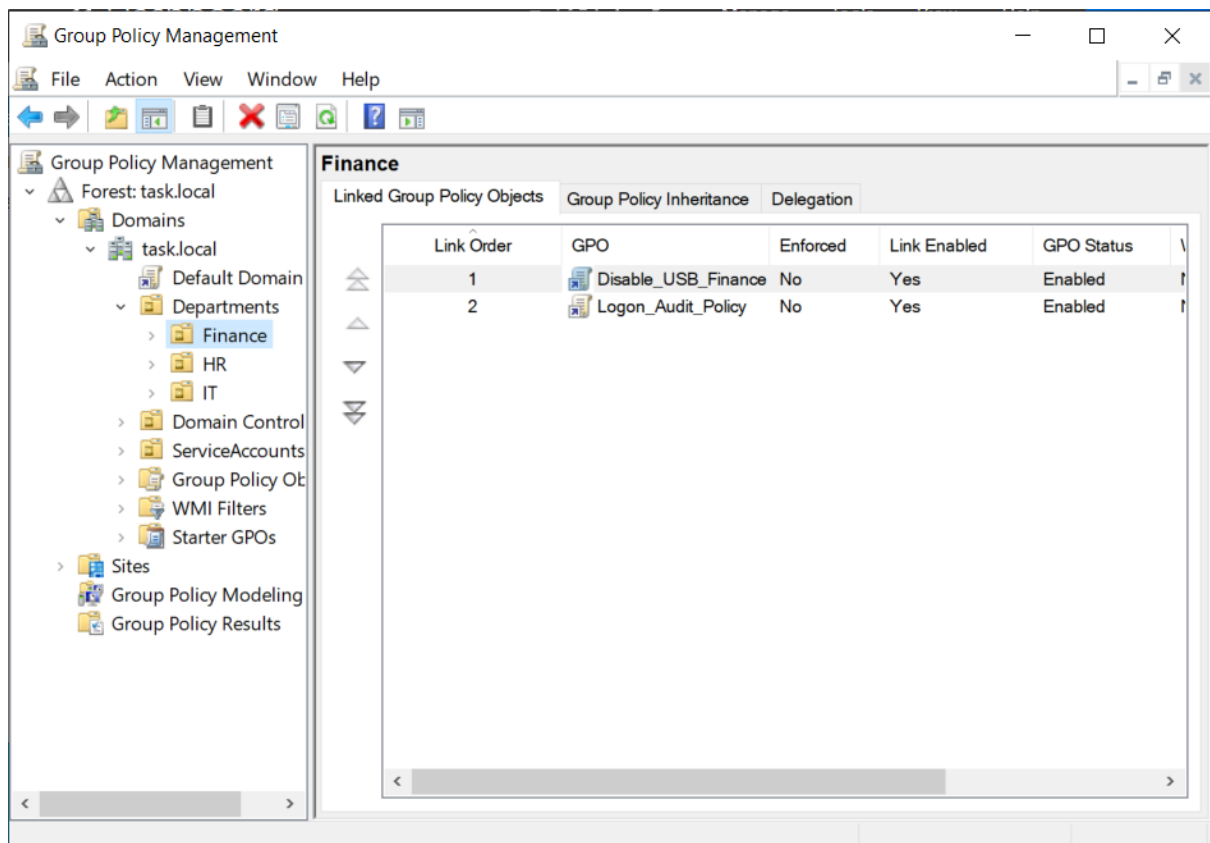
4. Methodology

4.1 Enable Audit Policy for Logon Events

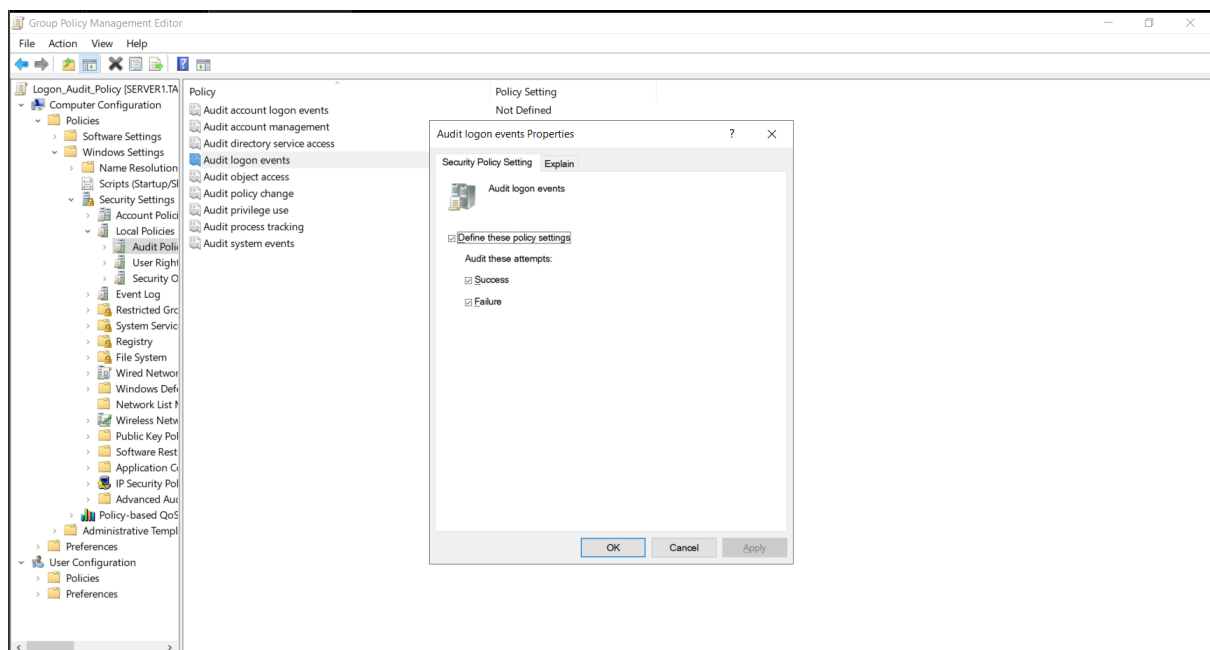
A GPO named Logon_Audit_Policy was created and linked to the Finance OU. In the GPO settings, "Audit logon events" was enabled for both Success and Failure under Audit Policy. The policy was updated on the client machine using gpupdate /force. Event Viewer on the client showed Event ID 4624 confirming successful logon logging.



(created a new gpo as Logon_Audit_Policy)



(Linked gpo with finance)



(Configure audit logon audit policy)

Administrator: Command Prompt

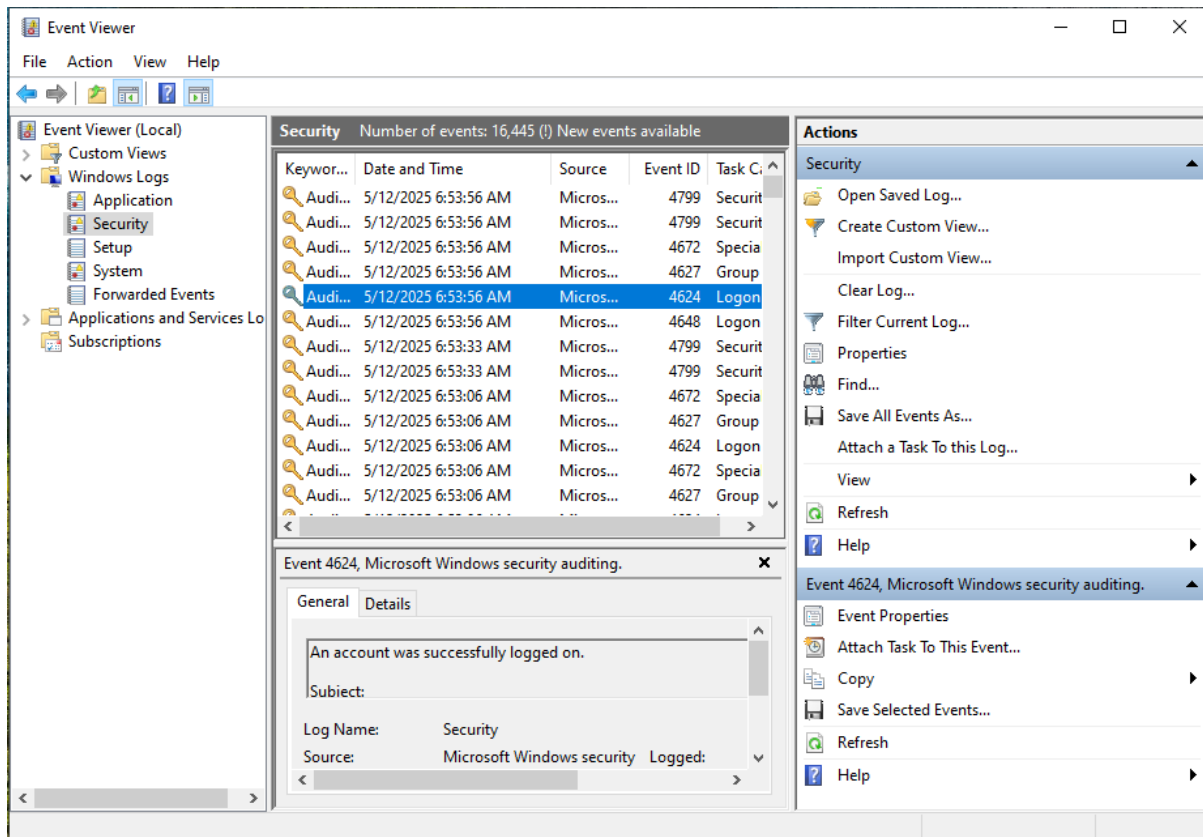
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Microsoft Windows [Version 10.0.19044.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Users\administrator>_
```

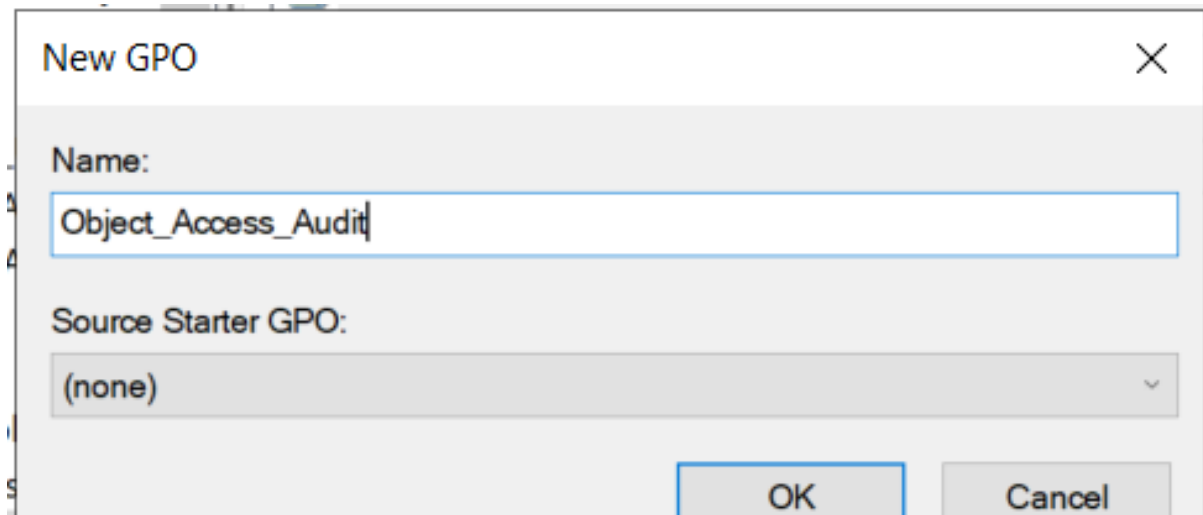
(Updated policy on pc1)



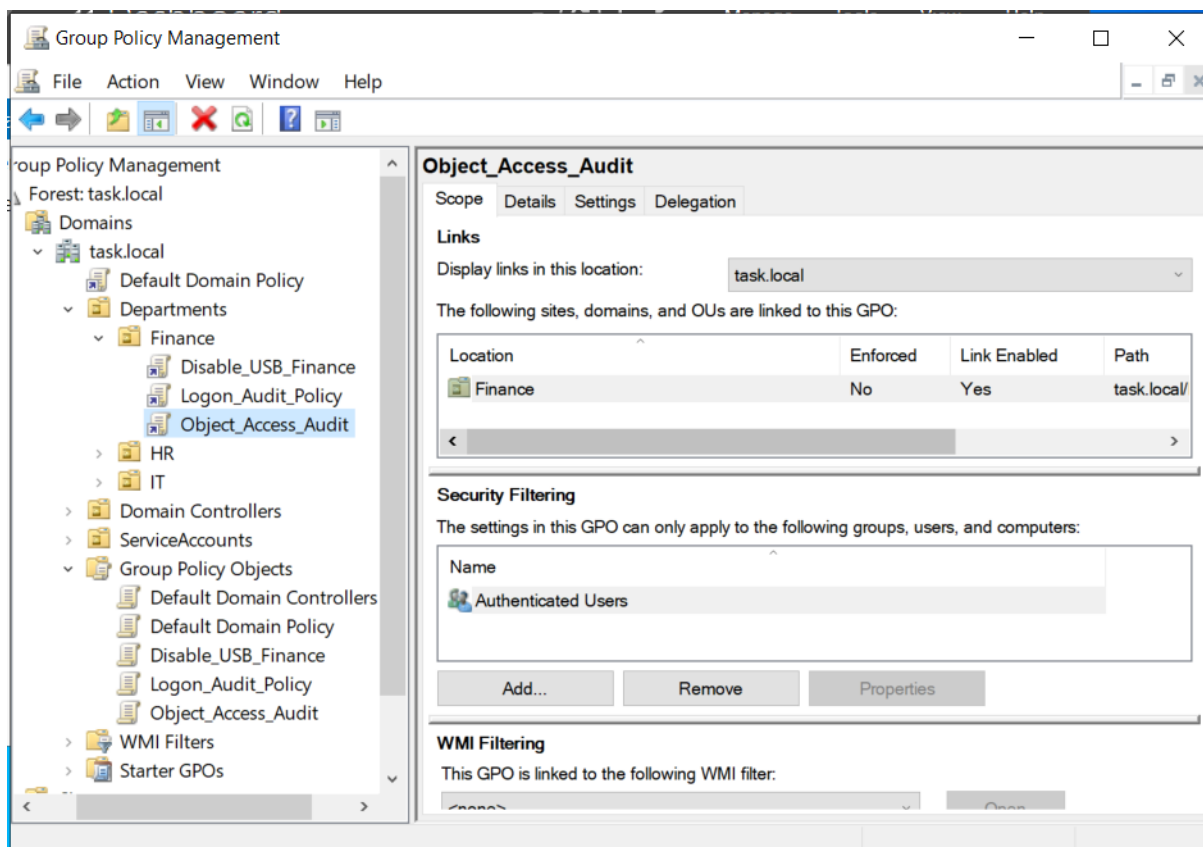
(Event ID 4624 – "An account was successfully logged on.")

4.2 Configure Audit Policy for Object Access

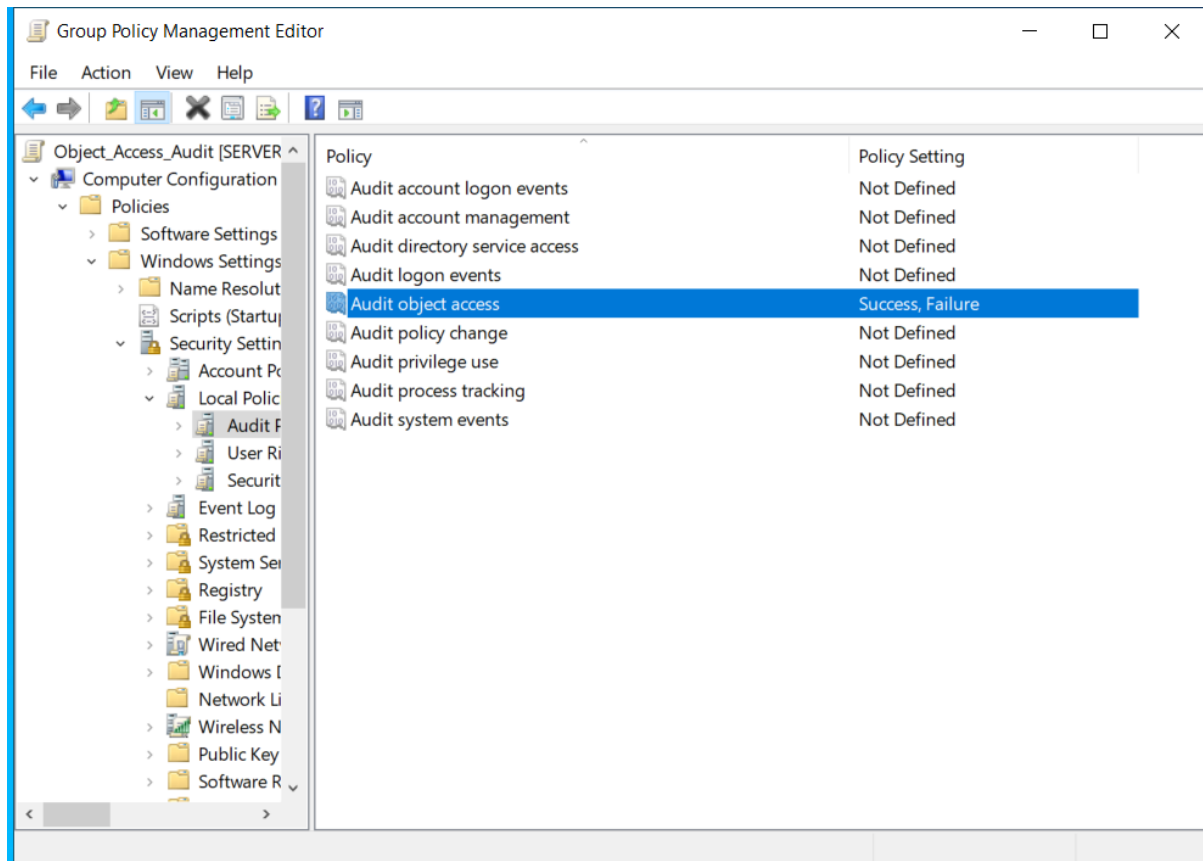
A GPO named Object_Access_Audit was created and linked to the Finance OU. "Audit object access" was enabled for Success and Failure. On the client, folder C:\AuditTest was created, and auditing was configured to log access by Everyone for Read, Write, and Delete. After interacting with the folder as a domain user, Event Viewer displayed Event ID 4663 confirming object access was logged.



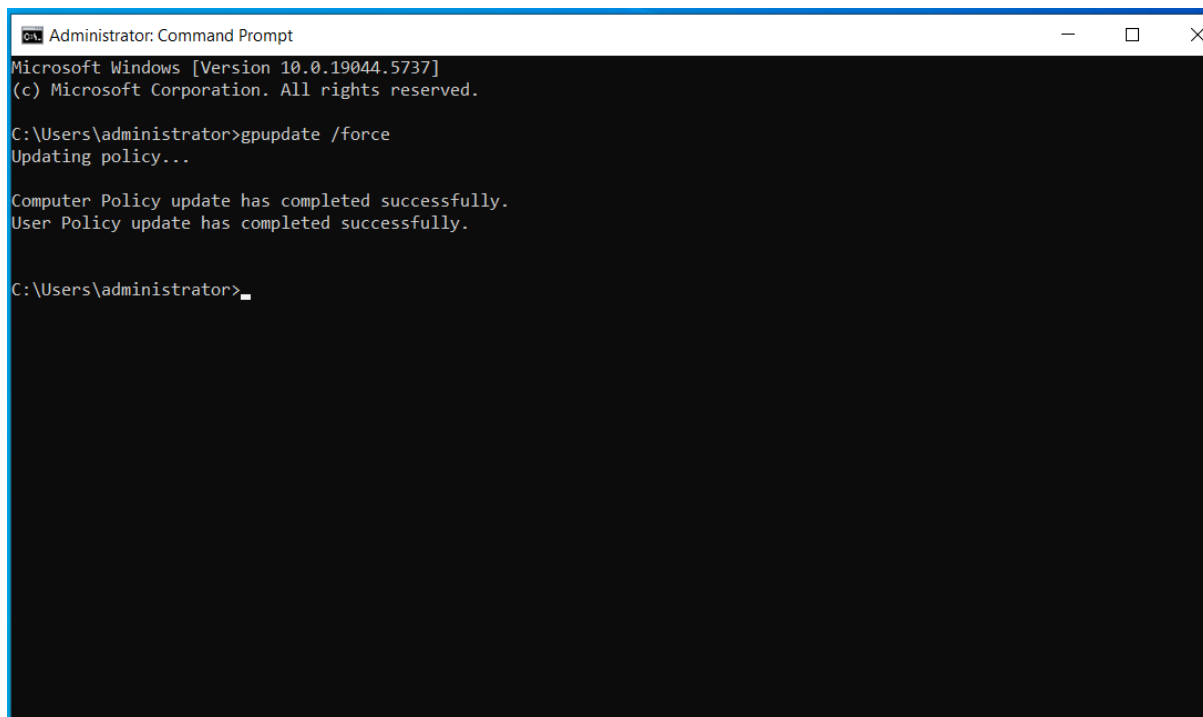
(Create a new gpo Object_Access_Audit)



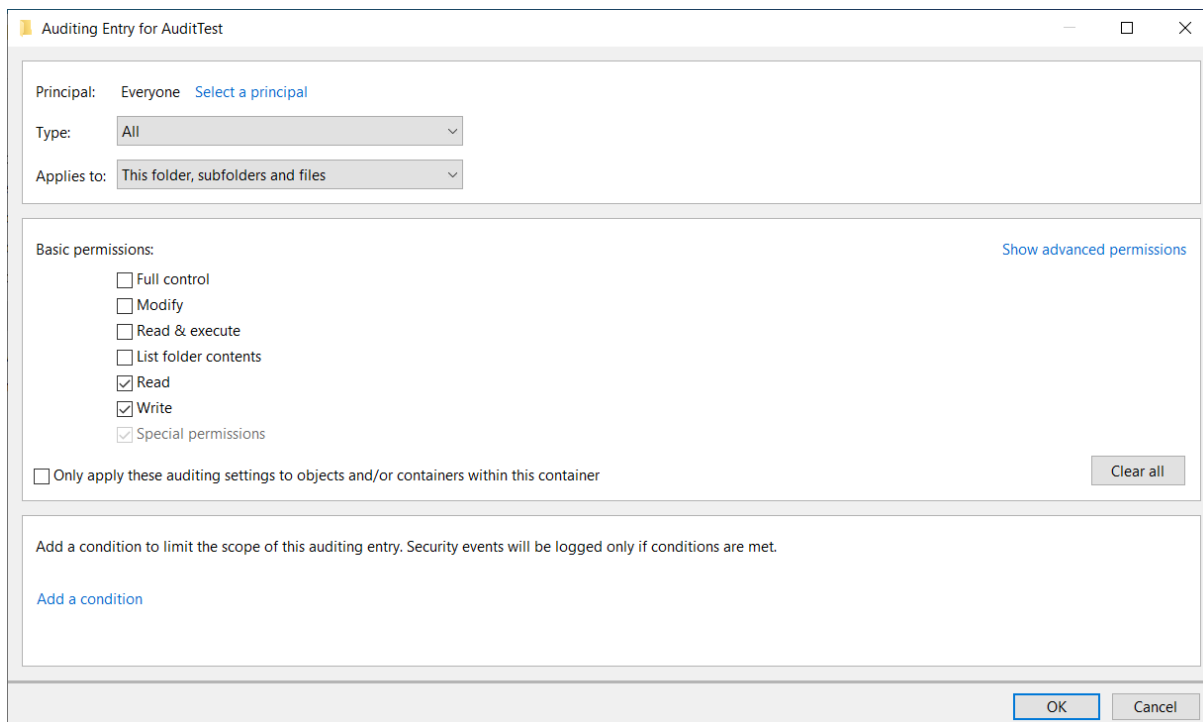
(Successfully created gpo and linked it with finance)



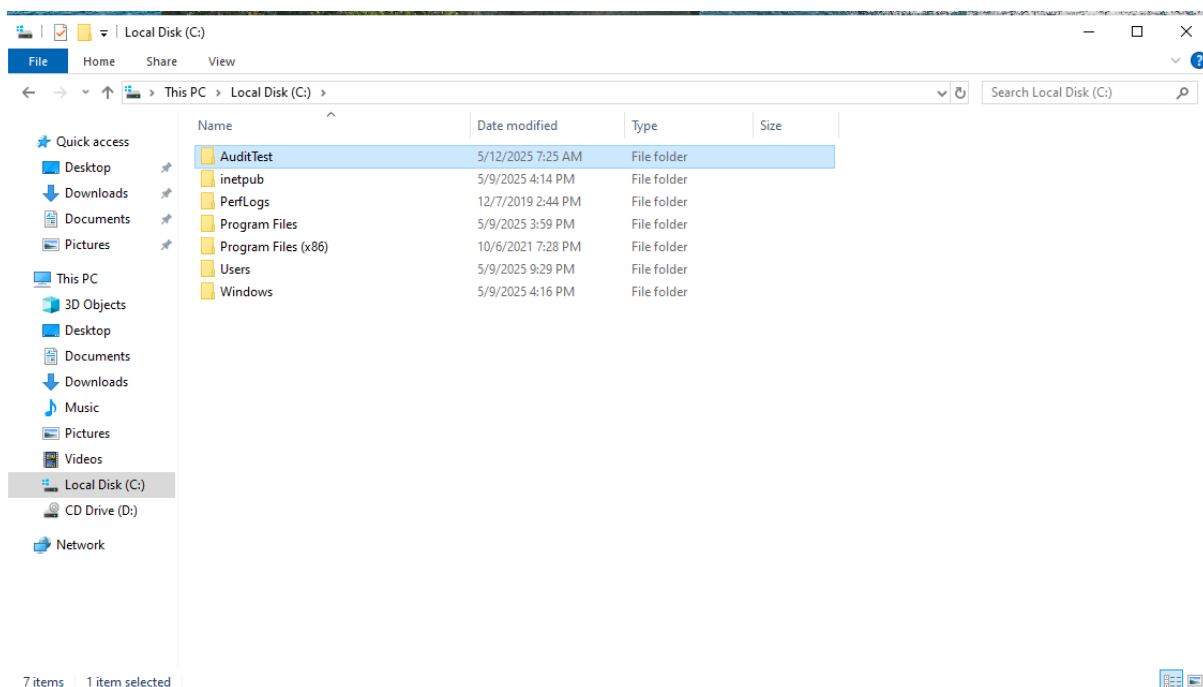
(Configured the policy)



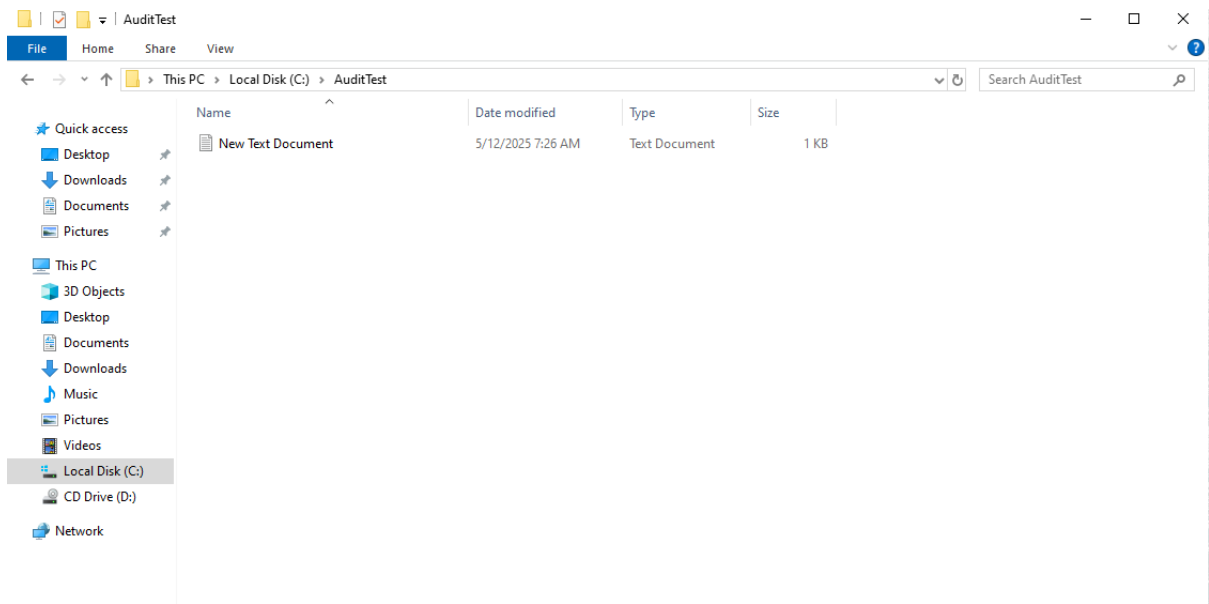
(Updated policy on PC1(domain pc))



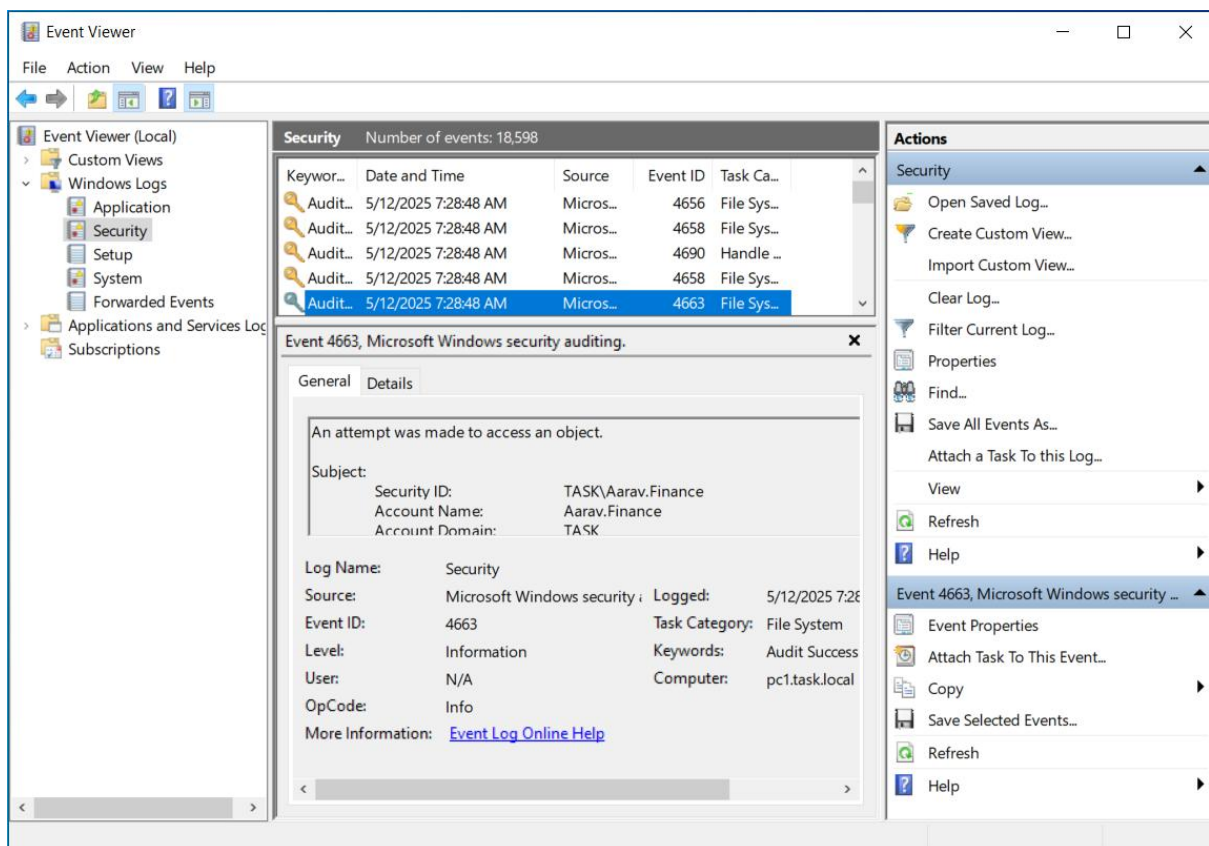
(edit the rule)



(Now audit folder is showing on user – aarav , which we created in admin account for testing purpose)



(Created a new file for testing audit rule)



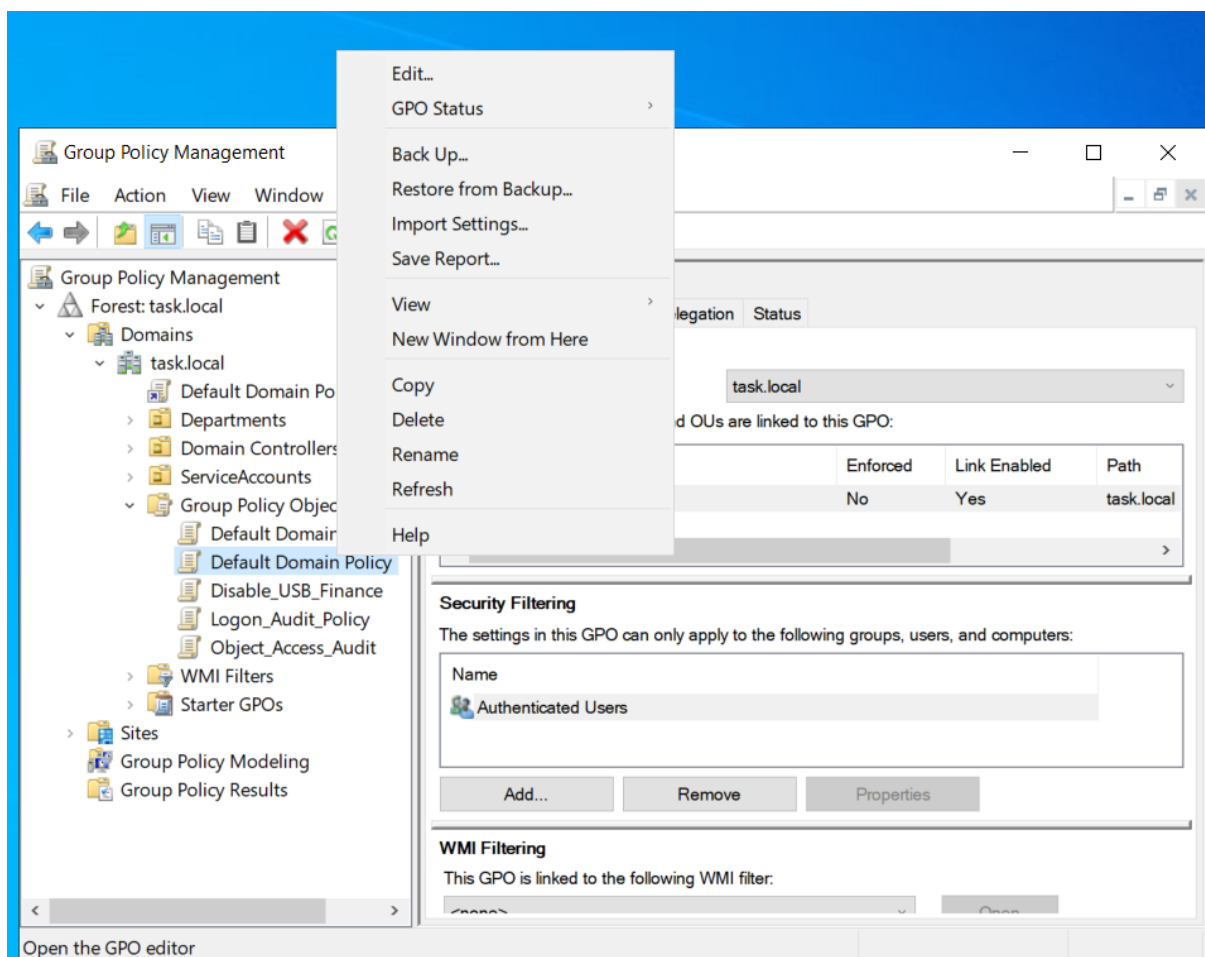
(In admin account of pc1 we can check Event ID 4663 showing access attempt)

4.3 Enable Account Lockout Policy

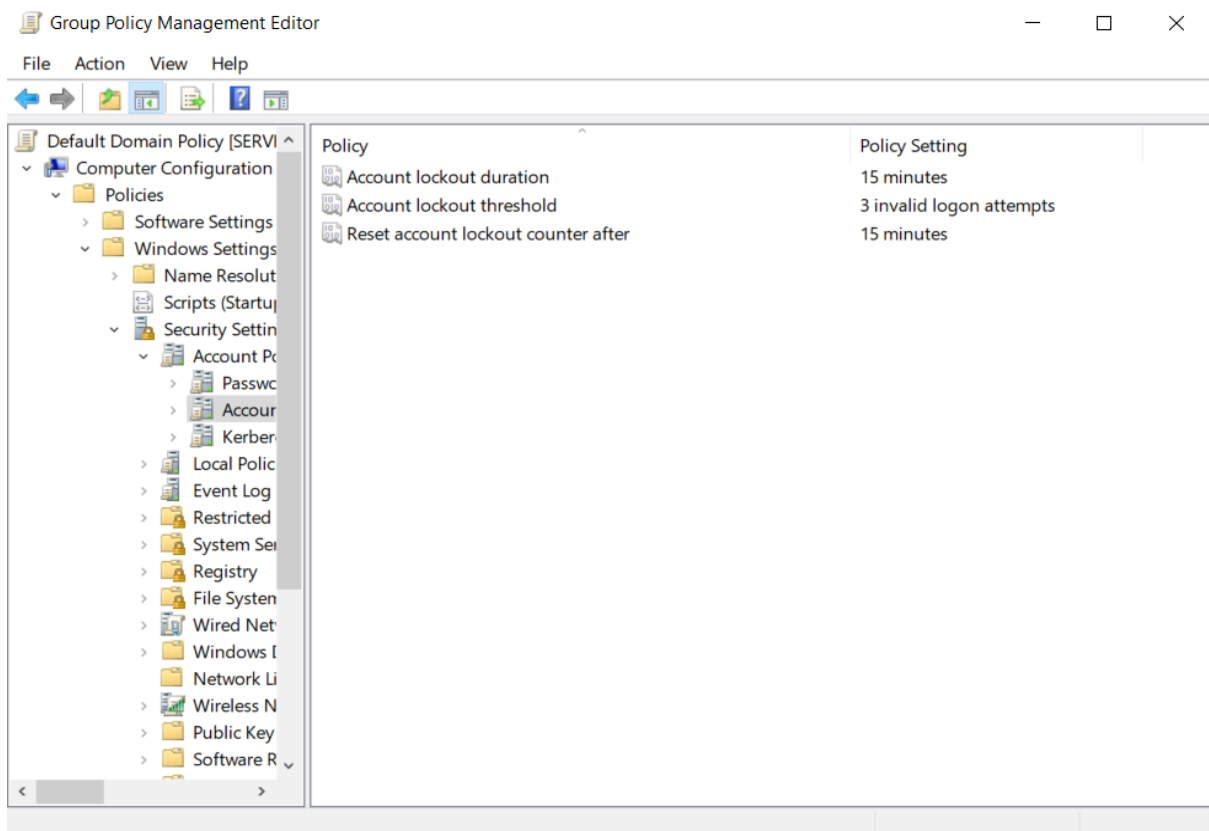
In the Default Domain Policy, the following settings were configured:

- Account lockout threshold: 3 invalid attempts
- Account lockout duration: 15 minutes
- Reset account lockout counter after: 15 minutes

After policy application, the user Aarav.Finance was locked out by entering the wrong password three times. The locked status was confirmed in ADUC under the Account tab.



(Edit default domain policy under group policy object)



(Editing account lockout policy)

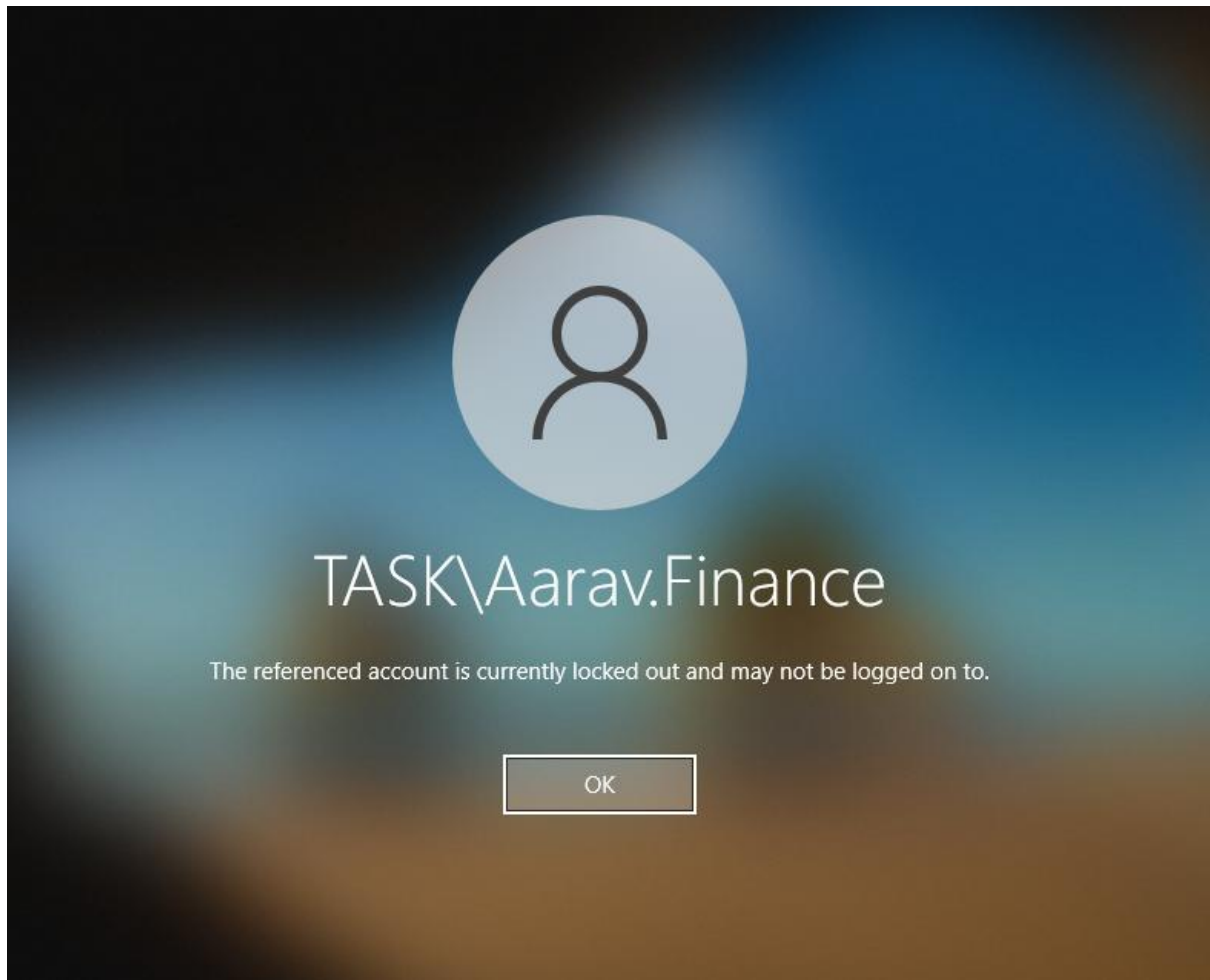
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19044.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>gpupdate /force
Updating policy...

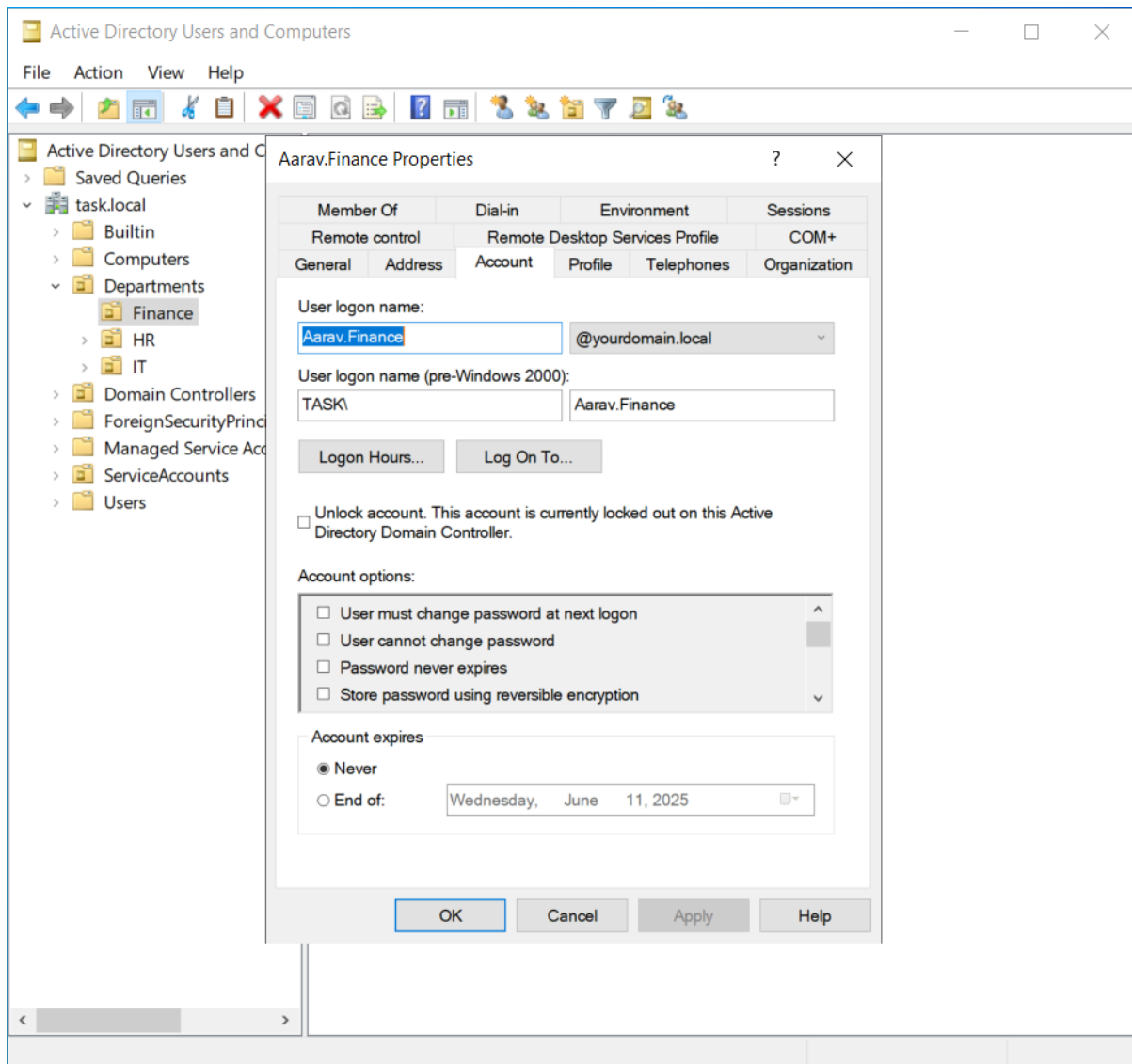
Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Users\administrator>
```

(Updated policy on pc1)



(After 3 incorrect passwords there is Lockout error message after 3rd attempt)



(Aarav.Finance is currently locked out on the domain controller, meaning the Account Lockout Policy is working as expected)

5. Recommendations

- Maintain logon auditing for all user accounts across the domain to track access patterns and support forensic investigations.
- Regularly review object access audit logs to detect unauthorized attempts to read, write, or delete sensitive files or directories.

- Avoid over-auditing system activity to prevent excessive log size or performance degradation on client machines.
- Configure lockout policies with realistic thresholds to balance security with usability and minimize lockout-related helpdesk incidents.
- Revisit and adjust audit policies periodically based on the organization's risk profile and any changes in regulatory or compliance requirements.

6. Conclusion

The configured logon auditing policy successfully recorded user authentication activity, and event logs verified the capture of successful login attempts. Object access auditing was properly set up through group policy and folder-level security settings, and file interaction was logged with Event ID 4663. The account lockout policy was enforced through changes to the Default Domain Policy and confirmed when a domain user account was locked after three incorrect login attempts. These controls establish essential logging capabilities and strengthen the overall security posture of the Active Directory environment by tracking user activity and responding to unauthorized access attempts.